

REVERSE ENGINEERING // MASTERY COURSE

PHASE 6

Specialization

Apna raasta chuno — malware, exploits, mobile, crypto, firmware.

```
re@learning: ~/phase-6

$ ./start_learning.sh --phase 6
# Hinglish mein, simple examples ke saath – ek-ek step
$ status: READY ✓
```



Detailed Theory



Code + Line-by-line



Interview Q&A

IN Hinglish

Reverse Engineering MasteryFA • Self-Study Roadmap 5 • Phase 6 Handbook

Phase 6 — Specialization (Apna Raasta Chuno)

Goal: Ab tak tum RE ki basics jaante ho. Ab decide karo kis direction mein deep jaana hai. RE bahut bada field hai — sab kuch expert nahi ban sakte, ek-do cheezon mein master bano. Time: **Long term.**

6.0 — Specialization Kyun?

Ab tak tum "generalist" ho — thoda sab kuch aata hai. Lekin career aur mastery ke liye ek direction chunni padti hai. Har specialization ka apna tools, techniques, aur job market hai.

Pehle 1-2 areas explore karo (jo interesting lage), phir usme deep jao. Zaroori nahi ek hi mein rehna, but ek time pe ek pe focus.

6.1 — Specialization Options

1. 🦟 Malware Analyst

Malware (virus, ransomware, trojans) ko analyze karna. - **Kya karte hain:** Malware kaise kaam karta, kaise phailta, kya chura raha — samajhna. Detection signatures banana. - **Skills:** Static + dynamic analysis, unpacking, Windows internals, network analysis. - **Career:** SOC analyst, threat researcher, incident responder. **Bahut demand.** - **Tools:** Ghidra, x64dbg, Cuckoo Sandbox, Wireshark, YARA.

2. 🐸 Vulnerability Researcher / Exploit Developer

Software mein bugs (kamzoriyan) dhoondna aur exploits banana. - **Kya karte hain:** Buffer overflows, use-after-free jaise bugs dhoondna; unhe exploit karna. - **Skills:** Deep assembly, memory corruption, fuzzing, exploit dev (ROP, shellcode). - **Career:** Bug bounty, red team, security research. **High paying.** - **Tools:** Fuzzers (AFL), pwntools, GDB, IDA.

3. 📱 Mobile RE (Android / iOS)

Mobile apps ko reverse karna. (Ye Phase 7 mein detail mein — poora plan hai.) - **Kya karte hain:** APK/IPA analyze, app security testing, protection bypass. - **Skills:** Smali (Android), ARM assembly, Frida, mobile frameworks. - **Career:** Mobile app pentester, security researcher.

4. 🎮 Game Hacking / Anti-Cheat RE

Games ke internals samajhna (learning ke liye). - **Kya karte hain:** Memory editing, cheats banana/samajhna, anti-cheat research. - **Skills:** Memory analysis, DirectX/OpenGL hooking, C++. - **Note:** Sirf learning/research ke liye — online games mein cheating ban/legal issues laa sakti hai.

5. 🗝️ Cryptography RE

Encryption algorithms ko samajhna/todna. - **Kya karte hain:** Custom crypto pehchanna, weak implementations dhoondna. - **Skills:** Math, crypto algorithms (AES, RSA), pattern recognition. - **Career:** Crypto researcher, security auditor.

6. ⚡ Firmware / IoT / Embedded RE

Hardware devices ka software (firmware) reverse karna. - **Kya karte hain:** Router, camera, smart devices ka firmware analyze. - **Skills:** ARM/MIPS assembly, firmware extraction, hardware interfaces (UART, JTAG). - **Career:** IoT security researcher, hardware hacker. - **Tools:** binwalk, Ghidra, hardware tools.

6.2 — Kaunsa Choose Karein? (Decision Helper)

Agar tumhe pasand hai...	Toh ye try karo
Virus/hacking news, defense	🦟 Malware Analyst
Bugs dhoondna, puzzles todna	🐛 Vuln Research
Mobile apps, Android/iOS	📱 Mobile RE
Games, memory tricks	🎮 Game Hacking
Math, encryption	🗝️ Crypto RE
Hardware, gadgets	⚡ Firmware/IoT

Job market ke hisaab se: **Malware Analysis** aur **Vulnerability Research** ki sabse zyada demand aur pay hai. But jo tumhe maza aaye wahi best hai — kyunki usme tum consistent rahoge.

6.3 — Advanced Topics (Jab Ready Ho)

Chahe koi bhi specialization chuno, ye advanced skills sabme kaam aati hain:

1. Packers & Unpacking

- Packed binaries (UPX, custom packers) ko unpack karna.

- Manual unpacking — OEP (Original Entry Point) dhoondna, memory dump lena.

2. Obfuscation & Deobfuscation

- Code ko jaan-boojh ke complex banana (obfuscation) aur use simplify karna.
- Control flow flattening, junk code, string encryption todna.

3. Exploit Development

- Buffer overflow → shellcode → ROP chains.
- Modern protections (ASLR, DEP, stack canaries) aur unhe bypass karna.

4. Automation & Scripting (★ Bahut important)

- **Python** seekho — RE mein automation ke liye must.
- **Ghidra scripting** — apne analysis tasks automate karna.
- **angr** — symbolic execution framework (automatic tarike se inputs solve karna).
- **Frida** — dynamic instrumentation (runtime hooking).

5. Kernel-level RE

- OS ke andar (drivers, rootkits) ka RE — advanced Windows/Linux internals.

6.4 — Career Path (RE Jobs)

RE skills se ye roles mil sakte hain: - **Malware Analyst / Reverse Engineer** — security companies, antivirus vendors. - **Vulnerability Researcher** — product security teams, bug bounty (freelance). - **Security Researcher** — R&D, zero-day research. - **Penetration Tester** — with RE as a specialty (mobile/binary). - **Threat Intelligence Analyst** — malware ko track karke reports banana.

Portfolio banao

- CTF achievements aur writeups.
- GitHub pe RE tools/scripts.
- Blog pe malware analysis / crackme writeups.
- Certifications (optional): GREM (malware), OSCP/OSED (exploit), etc.

🎯 Key Takeaways (Ek Line Mein)

1. RE bada field hai — ek-do areas mein **master** bano, sab mein nahi.
2. Options: **Malware, Vuln Research, Mobile, Game, Crypto, Firmware.**

3. Job demand + pay: **Malware Analysis** aur **Vuln Research** top pe.
4. Jo tumhe **maza** aaye wahi best — consistency wahin se aayegi.
5. Advanced skills sabme kaam: **unpacking, deobfuscation, exploit dev, scripting**.
6. **Python + Ghidra scripting + Frida + angr** — automation ke liye seekho.
7. **Portfolio** (writeups, CTF, GitHub, blog) career ke liye zaroori.

✓ **Phase 6 Ongoing:** Specialization ek journey hai, ek din ka kaam nahi. Ek area chuno, usme deep jao, projects banao. Aur agla logical step: **Mobile RE (Phase 7)** — poora detailed plan next folder mein. 🚀

Phase 6 — Interview Questions & Answers (Hinglish)

Specialization aur advanced topics se related important interview questions, simple Hinglish jawaab ke saath.

Q1. Reverse Engineering mein specialization kyun zaroori hai?

Ans: RE bahut bada field hai — malware, exploits, mobile, crypto, firmware sab alag areas hain, har ek ke apne tools aur techniques hain. Sab mein expert banna mushkil hai, isliye 1-2 areas chunkar unme deep jaana chahiye. Isse mastery aati hai aur career focused hota hai.

Q2. Malware Analyst kya karta hai?

Ans: Malware analyst virus, ransomware, trojans jaise malicious software ko analyze karta hai — samajhta hai wo kaise kaam karta, kaise phailta, kya chura raha hai. Fir detection signatures (jaise YARA rules) aur IOCs banata hai taaki systems ko bachaaya ja sake. Ye role security companies mein bahut demand mein hai.

Q3. Vulnerability researcher aur exploit developer kya karte hain?

Ans: Ye software mein bugs (kamzoriyan) dhoondte hain — jaise buffer overflow, use-after-free — aur unhe exploit karke dikhate hain ki wo kitne khatarnaak hain. Isse companies bugs fix kar paati hain.

Ismein deep assembly, memory corruption, aur exploit development (ROP, shellcode) ki samajh chahiye. Bug bounty aur red team mein ye skills kaam aati hain.

Q4. Packer kya hota hai aur unpacking kya hoti hai?

Ans: Packer ek tool hai jo program ke code ko compress ya encrypt kar deta hai (jaise UPX), taaki analysis mushkil ho aur size chhota ho. Unpacking matlab us packed code ko wapas asli form mein laana — usually program ko chala ke, jab wo khud ko memory mein unpack kar leta hai, tab OEP (Original Entry Point) pe memory dump le lete hain.

Q5. Obfuscation kya hoti hai?

Ans: Obfuscation matlab code ko jaan-boojh ke complex/confusing banana taaki reverse karna mushkil ho — bina behaviour badle. Techniques: junk (bekaar) code daalna, control flow flatten karna, strings encrypt karna, naam ulte-pulte karna. Deobfuscation ismein wo confusion hataake asli logic nikaalna hai.

Q6. Exploit development mein ASLR, DEP, stack canary kya hote hain?

Ans: Ye modern security protections hain jo exploits ko rokne hain: - **ASLR** — memory addresses ko har baar random kar deta hai, taaki attacker ko address pata na chale. - **DEP/NX** — data wale memory ko execute hone se rokta hai (shellcode na chale). - **Stack canary** — stack pe ek secret value rakhta hai; overflow se wo change ho jaaye toh program detect kar leta hai.

Exploit developers inhe bypass karne ke techniques (jaise ROP) seekhte hain.

Q7. RE mein scripting/automation kyun important hai?

Ans: Bahut sa RE kaam repetitive hota hai (patterns dhoondna, decrypt karna, inputs solve karna). Python jaise scripting se ye automate ho jaata hai. Ghidra scripting se analysis tasks, angr se automatic input solving (symbolic execution), aur Frida se runtime hooking automate hoti hai. Isse time bachta hai aur bade programs handle hote hain.

Q8. angr framework kya hai?

Ans: angr ek Python-based binary analysis framework hai jo "symbolic execution" karta hai — matlab program ko symbolically chalaakar automatically wo input dhoond leta hai jo kisi specific path (jaise "access granted") tak le jaaye. RE mein ise password/key automatically solve karne ke liye use karte hain.

Q9. Firmware RE aam software RE se kaise alag hai?

Ans: Firmware embedded devices (routers, cameras, IoT) ka software hota hai. Ye often ARM/MIPS architecture pe hota hai (x86 nahi), aur pehle device se firmware extract karna padta hai (binwalk jaise tools se). Kabhi hardware interfaces (UART, JTAG) se seedha chip se code nikalna padta hai. Baaki RE concepts same rehte hain.

Q10. YARA rules kya hote hain?

Ans: YARA rules ek tarah ke "patterns" hain jinse malware ko identify/detect kiya jaata hai. Rule mein hum specific strings, bytes, ya conditions likhte hain jo kisi malware family mein milte hain. Malware analysts ye rules banate hain taaki antivirus/scanners similar malware pakad sakein.

Q11. RE mein career ke kaunse options hain?

Ans: Malware Analyst, Reverse Engineer, Vulnerability Researcher, Security Researcher, Penetration Tester (binary/mobile specialty), aur Threat Intelligence Analyst. Ye roles security companies, antivirus vendors, product security teams, aur bug bounty (freelance) mein hote hain.

Q12. Specialization choose karte waqt kaunsi cheez sabse zyada matter karti hai?

Ans: Job demand aur pay important hain (Malware Analysis aur Vuln Research top pe hain), lekin sabse zyada matter karta hai ki tumhe kis cheez mein maza aata hai. Kyunki RE lambi mehnat maangti hai, aur consistency wahin aayegi jahan interest hoga. Interest + demand ka combo best hai.

Phase 6 — Interview Questions & Answers (English)

The same Specialization and advanced-topics interview questions, answered in clear English.

Q1. Why is specialization important in Reverse Engineering?

Ans: RE is a very broad field — malware, exploits, mobile, crypto, and firmware are all distinct areas with their own tools and techniques. Becoming expert in everything is impractical, so you pick 1-2 areas and go deep. This builds mastery and gives your career focus.

Q2. What does a Malware Analyst do?

Ans: A malware analyst examines malicious software like viruses, ransomware, and trojans — understanding how it works, spreads, and what it steals. They then create detection signatures (e.g., YARA rules) and IOCs to protect systems. The role is in high demand at security companies.

Q3. What do vulnerability researchers and exploit developers do?

Ans: They find bugs in software — such as buffer overflows and use-after-free — and build exploits to demonstrate their severity, so companies can fix them. It requires deep assembly, understanding of memory corruption, and exploit development (ROP, shellcode). These skills are used in bug bounties and red teaming.

Q4. What is a packer and what is unpacking?

Ans: A packer is a tool that compresses or encrypts a program's code (e.g., UPX) to make analysis harder and reduce size. Unpacking means restoring the original code — usually by running the program until it unpacks itself in memory, then dumping the memory at the OEP (Original Entry Point).

Q5. What is obfuscation?

Ans: Obfuscation is deliberately making code complex/confusing to hinder reversing, without changing its behavior. Techniques include inserting junk code, flattening control flow, encrypting strings, and mangling names. Deobfuscation removes that confusion to recover the real logic.

Q6. In exploit development, what are ASLR, DEP, and stack canaries?

Ans: These are modern protections that hinder exploits: - **ASLR** — randomizes memory addresses each run, so attackers can't predict them. - **DEP/NX** — marks data memory as non-executable, preventing shellcode from running. - **Stack canary** — places a secret value on the stack; if an overflow changes it, the program detects the tampering.

Exploit developers learn techniques (like ROP) to bypass these.

Q7. Why is scripting/automation important in RE?

Ans: Much RE work is repetitive (finding patterns, decrypting, solving inputs). Scripting with Python automates it. Ghidra scripting automates analysis tasks, angr automates input solving (symbolic execution), and Frida automates runtime hooking. This saves time and scales to large programs.

Q8. What is the angr framework?

Ans: angr is a Python-based binary analysis framework that performs "symbolic execution" — running a program symbolically to automatically find the input that reaches a specific path (like "access granted"). In RE it's used to automatically solve passwords/keys.

Q9. How is firmware RE different from ordinary software RE?

Ans: Firmware is the software of embedded devices (routers, cameras, IoT). It's often on ARM/MIPS architectures (not x86), and you must first extract the firmware from the device (with tools like binwalk). Sometimes you extract code directly from the chip via hardware interfaces (UART, JTAG). The core RE concepts remain the same.

Q10. What are YARA rules?

Ans: YARA rules are patterns used to identify/detect malware. In a rule you specify strings, bytes, or conditions characteristic of a malware family. Analysts write these rules so antivirus/scanners can catch similar malware.

Q11. What career options exist in RE?

Ans: Malware Analyst, Reverse Engineer, Vulnerability Researcher, Security Researcher, Penetration Tester (with a binary/mobile specialty), and Threat Intelligence Analyst. These roles exist at security companies, antivirus vendors, product security teams, and in bug bounty (freelance).

Q12. What matters most when choosing a specialization?

Ans: Job demand and pay matter (Malware Analysis and Vulnerability Research are top), but what matters most is what you genuinely enjoy. RE demands sustained effort, and consistency comes from interest. The best choice combines interest with demand.